WAVE INIT - Digital QR Payment System

Problem Statement

Current Challenges in Cable/WiFi Payment Collection

1. **Manual Payment Collection**
    - Traditional collection methods require physical visits
    - Time-consuming and labor-intensive process
    - High operational costs for bill collection
2. **Payment Tracking Issues**
    - Difficult to track pending payments
    - No centralized record of transactions
    - Manual bookkeeping prone to errors
3. **Customer Inconvenience**
    - Limited payment options for customers
    - No digital payment infrastructure
    - Inconvenient bill payment process
4. **Administrative Overhead**
    - Difficult to manage customer database
    - Manual reconciliation of payments
    - Lack of real-time payment status

Proposed Solution: WAVE INIT

WAVE INIT is a comprehensive digital QR-based payment system designed specifically for cable TV and WiFi service providers. It enables seamless digital payments through multiple channels including UPI, Cards, and Net Banking.

Key Features

For Customers

- **Multiple Payment Options**: UPI (GPay, PhonePe, Paytm, BHIM), Debit/Credit Cards, Net Banking
- **QR Code Payments**: Scan and pay instantly
- **Mobile-First Design**: Accessible from any device
- **Real-time Payment Confirmation**: Instant transaction verification

For Administrators

- **Customer Management**: Add, edit, delete customer records

- **Dashboard Analytics**: View total customers, active users, pending payments
- **Transaction History**: Complete payment records
- **Pending Payment Tracking**: Monitor and follow up on unpaid bills

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Technology Stack

Component	Technology
Frontend	React.js
Backend	Spring Boot (Java)
Database	MySQL
Payment Integration	UPI, Card, Net Banking
API Architecture	RESTful

System Architecture

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│   Mobile/PC      │──▶│   React UI       │──▶│  Spring Boot     │
│   (Port 4000)    │     │   (Frontend)     │     │   (Port 5555)    │
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                                                  ┌──────────────────┐
                                                  │      MySQL       │
                                                  │    Database      │
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Payment Flow

1. Customer logs in with Customer ID
2. Views pending amount and bill details
3. Selects payment method (UPI/Card/Net Banking)
4. For UPI: Opens app or scans QR code
5. Payment verification
6. Confirmation and receipt

UPI Integration

- **Merchant UPI ID**: titooram123@oksbi
- **Supported Apps**: Google Pay, PhonePe, Paytm, BHIM
- **QR Code Generation**: Dynamic QR for each transaction
- **Payment Verification**: Real-time confirmation

Benefits

For Service Provider

- ☑ Reduced collection costs
- ☑ Faster payment reconciliation
- ☑ Better customer management
- ☑ Real-time financial insights

For Customers

- ☑ Convenient digital payments
- ☑ Multiple payment options
- ☑ 24/7 payment access
- ☑ Instant payment confirmation

Security Features

- Secure authentication system
- Transaction verification
- Encrypted data transmission
- Role-based access control (Admin/Customer)

Conclusion

WAVE INIT transforms traditional payment collection into a modern, efficient digital solution. By implementing this system, service providers can significantly reduce operational costs while providing customers with a seamless payment experience.

Organization: WAVE INIT
Version: 1.0
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This document contains confidential information about the WAVE INIT payment system.